

CORE PLATFORM FRAMEWORK

91 Gateway 매뉴얼

Route · Target · Security · Resilience · Publish · LKG/Rollback · Reconcile

항목	기준
대상	Gateway/API 운영자, 플랫폼 운영자, 보안 운영자
정보 역할	외부 진입 Gateway 사용·운영 정보
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요구사항 정보	CPF_FINAL_TARGET_REQUIREMENTS.md · 180 개

이 문서는 현재 구현의 상태 보고가 아니라, CPF 제품 사용자가 업무를 끝내기 위해 따라야 할 완성 기준 사용 계약을 중심으로 작성한다. 실제 CPF Symbol/Path 는 CURRENT SOURCE, 구현 독립 예시는 REFERENCE 로 구분한다.

문서 읽는 법과 사실 구분

표기	의미	사용 규칙
CURRENT SOURCE	최신 master 에서 직접 확인한 실제 Class/API/Route/Property/SQL/Test/경로	복사·검색 가능한 실제 이름으로 사용한다.
PRODUCT CONTRACT	CPF 가 완료된 제품으로서 제공해야 하는 사용·운영 계약	현재 구현량 때문에 축소하지 않는다.
REFERENCE	사용자가 그대로 응용할 수 있도록 제공하는 완성 기준 코드·설정·절차 예시	실제 CPF 내부 Symbol 인 것처럼 표현하지 않는다.

본문은 기능을 찾는 데서 끝나지 않는다. 적용 가능한 범위에서 선택 기준 → 준비 → 구현/설정 → 실행 → 정상 결과 → 실패/부분 실패 → 복구·대사 → 보안·감사 → Test → 운영 인계 순서로 설명한다.

1. Gateway 를 언제 쓰는가

사용	판단
외부 Client→내부 API trust boundary	Gateway 사용
인증/허용 Client/route/rate/TLS 정책 집행	Gateway 사용
내부 Domain A→B 호출	Gateway 재경유 금지; Local/Remote Facade 사용
Batch→Domain 내부 호출	Gateway 재경유 금지
단순 reverse proxy 만 필요	CPF Gateway 의 운영/ledger/approval 가치와 비교 후 결정

2. 설치와 현재 기본 Config

[CURRENT SOURCE] cpf-gateway/src/main/resources/application.yml

Property	Default	운영 의미
GWY_SERVER_PORT	8070	Data plane port
GWY_ROUTE_REFRESH	30s	Route refresh
GWY_POLICY_REFRESH	15s	Policy refresh
GWY_HEALTH_PROBE_INTERVAL	30s	Target probe
GWY_HEALTH_STALE_AFTER	90s	stale threshold
GWY_CONNECT_TIMEOUT_CAP	10s	connect timeout cap
GWY_RESPONSE_TIMEOUT_CAP	60s	response cap
GWY_OVERALL_TIMEOUT_CAP	90s	overall cap
GWY_RETRY_COUNT_CAP	2	retry upper bound
GWY_RATE_LIMIT_COUNTER_ENTRIES_CAP	100000	counter bound
GWY_RATE_LIMIT_COUNTER_MODE	LOCAL	counter mode
GWY_RATE_LIMIT_FAIL_CLOSED	true	counter failure policy
GWY_REQUIRE_TLS_INGRESS	false	ingress TLS policy
GWY_REQUEST_BODY_BYTES_CAP	10485760	request size cap
GWY_RESPONSE_BODY_BYTES_CAP	10485760	response size cap
GWY_HEADER_COUNT_CAP	100	header count cap
GWY_HEADER_BYTES_CAP	65536	header bytes cap
GWY_RAW_BODY_CAPTURE_ALLOWED	false	raw capture guard
GWY_LOG_SPOOL_BYTES_CAP	2147483648	spool bound
GWY_BOOTSTRAP_MODE	FAIL_CLOSED	invalid/missing snapshot policy

Property	Default	운영 의미
GWY_CONTROL_ENABLED	false	control listener
GWY_CONTROL_ALLOWED_SKEW_SECONDS	60	HMAC timestamp skew
GWY_CONTROL_NONCE_RETENTION_SECONDS	180	nonce replay window
GWY_CONTROL_BIND_ADDRESS	127.0.0.1	control bind
GWY_ROUTE_RECONCILE_MILLIS	30000	desired/actual reconcile

3. Route·Predicate·Filter·Rewrite

Route Reference

```
Route Definition (REFERENCE)
routeId: PAYMENTS_V1
version: 7
predicates:
  - host: api.partner.example
  - path: /payments/**
  - method: [POST, GET]
filters:
  - trustedHeaderNormalize
  - authentication
  - authorization
  - rateLimit
  - timeoutBudget
rewrite:
  stripPrefix: /payments
  targetPath: /internal/payments
```

Route snapshot 은 검증된 version/checksum 으로 atomic publish 한다. invalid/stale snapshot 을 부분 적용하지 않는다. 요청 URL 의 scheme/host/port/path 는 SSRF allowlist 와 canonicalization 을 통과한다.

4. Target·Discovery·Load Balancing

항목	계약
Target	service/instance/zone/version/weight/health/maintenance
Discovery	registry TTL/lease/state cleanup
Load balance	weight/zone/version/health 기반 결정적 선택
Failover	overall deadline 과 idempotency/replay safety 안에서만
Draining	신규 요청 제외, in-flight 종료 관측
Multi-instance	동일 route version/checksum desired state 공유

5. Authentication·Authorization·HMAC

Control HMAC 검증 의미

```
canonical = method + "
" + path + "
" + timestamp + "
" + nonce
          + "
" + audience + "
" + bodySha256;
signature = HMAC_SHA256(sharedSecret, canonical);

verify:
1. caller/audience allowlist
2. timestamp skew <= 60s (default control plane)
3. nonce not replayed within retention window
4. body hash matches bytes actually processed
5. constant-time signature compare
```

5.1 SSRF/TLS

- scheme/host/port/CIDR/service allowlist
- URI canonicalization 과 redirect 정책
- DNS rebinding/resolve policy
- TLS protocol/certificate/hostname/key rotation
- trusted proxy/header chain 과 내부 Header spoof 차단

6. Timeout·Retry·Circuit Breaker·Bulkhead

단계	현재 상한/정책
connect	10s cap
response	60s cap
overall	90s cap
retry	count cap 2
request/response body	10MB default cap
one-shot/stream body	안전 replay 가 없으면 retry 금지
rate limit	counter failure 기본 fail-closed
bulkhead	route/target 별 concurrency/resource bound

Retry 횟수보다 중요한 것은 “요청이 재생 가능한가”다. **body streaming/외부 side effect/response loss** 에서 결과가 불명확하면 **Attempt Ledger** 와 결과 조회를 사용한다.

7. Idempotency·Attempt Ledger·UNKNOWN_RESULT

PRODUCT CONTRACT - Attempt Ledger

```
AttemptLedger {  
  routeId, requestId, idempotencyKey, transactionId,  
  attempt, targetInstance, requestHash,  
  sendState, responseState, httpStatus,  
  failureStage, startedAt, completedAt,  
  outcome = SUCCESS | FAILED | UNKNOWN  
}
```

Failure stage	판정
before connect	FAILED/retryable 가능
connect/send partial	UNKNOWN 가능
request sent, response timeout	UNKNOWN
response received	SUCCESS/FAILED 확정
client disconnect after target side effect	원 target outcome 과 client delivery outcome 분리

8. Validation·Version·Checksum·Approval·Publish

1. Draft Route/Binding 작성
2. Schema/target/security/SSRF/timeout/retry validation
3. version + canonical checksum 생성
4. 고위험 변경은 Preview/Reason/Approval/command hash
5. Publish desired snapshot
6. Gateway instance ACK/NACK/PARTIAL 수집
7. actual version/checksum drift 확인
8. LKG promote 또는 rollback
9. audit + deployment timeline 종료

9. ACK/NACK/PARTIAL·LKG·Rollback

상태	의미	행동
ACK	instance 가 해당 version/checksum 활성화	전체 target ACK/정책 quorum 충족 여부 확인
NACK	validation/apply 실패	원인 격리; 해당 instance 는 이전 LKG 유지
PARTIAL	일부 instance 만 적용	신규 rollout 중지; desired/actual 대사; 선택 rollback/reapply
UNKNOWN	control response loss	instance actual snapshot 재조회
LKG	마지막 승인/검증된 snapshot	rollback source

10. Scale-out·Drift·Reconciliation

새 Gateway instance 는 임의 local config 가 아니라 승인된 desired snapshot 을 받아 checksum 을 검증한 뒤 ready 가 된다. desired/actual mismatch, stale instance, route event loss 는 주기적 reconciliation 으로 복구하며, 운영자는 apply-status 에서 version/checksum/instance 별 상태를 확인한다.

11. ADM Gateway Route 운영

ADM Route	화면	Risk	운영
/gateway-dashboard	Gateway 대시보드	MEDIUM	capability/snapshot/events/stream
/gateway-servers	Gateway 연동 서버	MEDIUM	server groups/members/save/delete
/gateway-groups	Gateway 서버 그룹	MEDIUM	groups/members/save/delete
/gateway-routes	Gateway 경로 · 라우팅	MEDIUM	bindings/save/state/delete
/gateway-security	Gateway 보안 · 제한	HIGH	bindings/security/state
/gateway-health	Gateway Health · 연결시험	MEDIUM	apply status/test/cancel/revalidate
/gateway-transactions	Gateway 거래 조회	MEDIUM	snapshot/events/trace
/gateway-log-policies	Gateway 로그 정책	MEDIUM	snapshot/log policy/distribution
/gateway-apply-status	Gateway 적용 상태 · 이력	MEDIUM	apply/events/test operation

12. 장애 Runbook

Target Down

10. Health/connection test 로 target 상태 확인
11. circuit/retry storm 차단
12. route alternate target 정책 확인
13. in-flight UNKNOWN ledger 추출
14. 복구 후 probe + ledger reconcile

Publish Partial

15. desired version/checksum 고정
16. instance ACK/NACK/UNKNOWN 분류
17. 신규 publish 중지
18. LKG rollback 또는 failed instance reapply
19. all instance actual state 대사

Control HMAC Failure

- 20. clock skew/nonce/audience/body hash/secret version 확인
- 21. replay/forgery 여부 audit
- 22. secret rotate 가 필요하면 dual control
- 23. 정상 control probe

Rate Counter Failure

- 24. counter mode/health 확인
- 25. fail-closed 정책에 따른 traffic 영향 판단
- 26. capacity/alternate distributed store
- 27. counter 복구 후 drift 확인

Spool/Disk Full

- 28. spool cap/disk 확인
- 29. raw capture disabled 여부
- 30. 업무 traffic 보호 정책 적용
- 31. log durability 복구/quarantine/replay
- 32. terminal loss 확인

13. EDU

13.1 Route publish

정상 경로를 먼저 확인한 뒤 Fault Injection: instance 하나 NACK. transactionId/requestId/attempt/version/checksum 을 기록하고, 결과가 불명하면 actual target/instance 상태를 조회해 reconcile 한 뒤 종료한다.

13.2 Response loss

정상 경로를 먼저 확인한 뒤 Fault Injection: target 처리 후 Gateway response timeout. transactionId/requestId/attempt/version/checksum 을 기록하고, 결과가 불명하면 actual target/instance 상태를 조회해 reconcile 한 뒤 종료한다.

13.3 SSRF

정상 경로를 먼저 확인한 뒤 Fault Injection: private/loopback forbidden target. transactionId/requestId/attempt/version/checksum 을 기록하고, 결과가 불명하면 actual target/instance 상태를 조회해 reconcile 한 뒤 종료한다.

13.4 Streaming retry

정상 경로를 먼저 확인한 뒤 Fault Injection: non-replayable body. transactionId/requestId/attempt/version/checksum 을 기록하고, 결과가 불명하면 actual target/instance 상태를 조회해 reconcile 한 뒤 종료한다.

13.5 Scale-out

정상 경로를 먼저 확인한 뒤 Fault Injection: 새 instance stale snapshot. transactionId/requestId/attempt/version/checksum 을 기록하고, 결과가 불명하면 actual target/instance 상태를 조회해 reconcile 한 뒤 종료한다.

13.6 LKG rollback

정상 경로를 먼저 확인한 뒤 Fault Injection: partial apply. transactionId/requestId/attempt/version/checksum 을 기록하고, 결과가 불명하면 actual target/instance 상태를 조회해 reconcile 한 뒤 종료한다.

13.7 Control HMAC

정상 경로를 먼저 확인한 뒤 Fault Injection: nonce replay. transactionId/requestId/attempt/version/checksum 을 기록하고, 결과가 불명하면 actual target/instance 상태를 조회해 reconcile 한 뒤 종료한다.

13.8 TLS rotation

정상 경로를 먼저 확인한 뒤 Fault Injection: old/new cert overlap. transactionId/requestId/attempt/version/checksum 을 기록하고, 결과가 불명하면 actual target/instance 상태를 조회해 reconcile 한 뒤 종료한다.

Source Trace

구분	Repository 경로	확인 포인트
Config	cpf-gateway/src/main/resources/application.yml	port/timeouts/rate/TLS/body/spool/control/reconcile
ADM Routes	cpf-admin/frontend/src/app/routes.ts	gateway-* operations
Requirement	CPF_FINAL_TARGET_REQUIREMENTS.md	Gateway/External/Event
Current Request	cpf-docs/work/CPF_CURRENT_WORK_REQUEST.md	transaction lineage/new reliability

14. Gateway 구성·운영 Recipe

14.1 신규 Route

server group→binding→security→timeout/rate→validate→approve→publish

Gate	판정
Preview	route/target/security/timeout/retry/rate/checksum 영향
Security	authn/authz/trusted header/SSRF/TLS/body policy
Reliability	idempotency/replay safety/attempt ledger/unknown
Publish	version/checksum + approval + target instances
Observe	ACK/NACK/UNKNOWN + request metrics
Failure	partial/response loss/target failure 를 상태별로 분리
Recover	LKG/actual state/result query/reconcile
Audit	requester/approver/reason/version/checksum/result

14.2 Target 교체

new target health→weight 0→canary→promote→old drain

Gate	판정
Preview	route/target/security/timeout/retry/rate/checksum 영향
Security	authn/authz/trusted header/SSRF/TLS/body policy
Reliability	idempotency/replay safety/attempt ledger/unknown
Publish	version/checksum + approval + target instances
Observe	ACK/NACK/UNKNOWN + request metrics
Failure	partial/response loss/target failure 를 상태별로 분리
Recover	LKG/actual state/result query/reconcile
Audit	requester/approver/reason/version/checksum/result

14.3 TLS Rotation

new cert/keyId preload→dual trust→switch→old revoke

Gate	판정
Preview	route/target/security/timeout/retry/rate/checksum 영향
Security	authn/authz/trusted header/SSRF/TLS/body policy
Reliability	idempotency/replay safety/attempt ledger/unknown
Publish	version/checksum + approval + target instances
Observe	ACK/NACK/UNKNOWN + request metrics
Failure	partial/response loss/target failure 를 상태별로 분리
Recover	LKG/actual state/result query/reconcile
Audit	requester/approver/reason/version/checksum/result

14.4 SSRF 정책

allow scheme/host/port/CIDR/service→negative test→publish

Gate	판정
Preview	route/target/security/timeout/retry/rate/checksum 영향
Security	authn/authz/trusted header/SSRF/TLS/body policy
Reliability	idempotency/replay safety/attempt ledger/unknown
Publish	version/checksum + approval + target instances
Observe	ACK/NACK/UNKNOWN + request metrics
Failure	partial/response loss/target failure 를 상태별로 분리
Recover	LKG/actual state/result query/reconcile
Audit	requester/approver/reason/version/checksum/result

14.5 Rate Limit

scope/key/window/cap + distributed counter requirement

Gate	판정
Preview	route/target/security/timeout/retry/rate/checksum 영향
Security	authn/authz/trusted header/SSRF/TLS/body policy
Reliability	idempotency/replay safety/attempt ledger/unknown
Publish	version/checksum + approval + target instances
Observe	ACK/NACK/UNKNOWN + request metrics
Failure	partial/response loss/target failure 를 상태별로 분리
Recover	LKG/actual state/result query/reconcile
Audit	requester/approver/reason/version/checksum/result

14.6 Streaming Upload

non-replayable body, bounded transfer/temp cleanup

Gate	판정
Preview	route/target/security/timeout/retry/rate/checksum 영향
Security	authn/authz/trusted header/SSRF/TLS/body policy
Reliability	idempotency/replay safety/attempt ledger/unknown
Publish	version/checksum + approval + target instances
Observe	ACK/NACK/UNKNOWN + request metrics
Failure	partial/response loss/target failure 를 상태별로 분리
Recover	LKG/actual state/result query/reconcile
Audit	requester/approver/reason/version/checksum/result

14.7 Response Loss

ledger UNKNOWN→target result query

Gate	판정
Preview	route/target/security/timeout/retry/rate/checksum 영향
Security	authn/authz/trusted header/SSRF/TLS/body policy
Reliability	idempotency/replay safety/attempt ledger/unknown
Publish	version/checksum + approval + target instances
Observe	ACK/NACK/UNKNOWN + request metrics
Failure	partial/response loss/target failure 를 상태별로 분리
Recover	LKG/actual state/result query/reconcile
Audit	requester/approver/reason/version/checksum/result

14.8 Partial Apply

desired snapshot vs instance actual checksum→LKG rollback/reapply

Gate	판정
Preview	route/target/security/timeout/retry/rate/checksum 영향
Security	authn/authz/trusted header/SSRF/TLS/body policy
Reliability	idempotency/replay safety/attempt ledger/unknown
Publish	version/checksum + approval + target instances
Observe	ACK/NACK/UNKNOWN + request metrics
Failure	partial/response loss/target failure 를 상태별로 분리
Recover	LKG/actual state/result query/reconcile
Audit	requester/approver/reason/version/checksum/result

14.9 Scale-out

new instance bootstrap desired snapshot→checksum→ready

Gate	판정
Preview	route/target/security/timeout/retry/rate/checksum 영향
Security	authn/authz/trusted header/SSRF/TLS/body policy
Reliability	idempotency/replay safety/attempt ledger/unknown
Publish	version/checksum + approval + target instances
Observe	ACK/NACK/UNKNOWN + request metrics
Failure	partial/response loss/target failure 를 상태별로 분리
Recover	LKG/actual state/result query/reconcile
Audit	requester/approver/reason/version/checksum/result

14.10 Control Secret Rotation

new secret/keyId/overlap→HMAC probe→old revoke

Gate	판정
Preview	route/target/security/timeout/retry/rate/checksum 영향
Security	authn/authz/trusted header/SSRF/TLS/body policy
Reliability	idempotency/replay safety/attempt ledger/unknown
Publish	version/checksum + approval + target instances
Observe	ACK/NACK/UNKNOWN + request metrics
Failure	partial/response loss/target failure 를 상태별로 분리
Recover	LKG/actual state/result query/reconcile
Audit	requester/approver/reason/version/checksum/result

15. ADM Gateway 화면별 절차

15.1 /gateway-dashboard

항목	내용
목적	관제 snapshot/event/stream
검색	route/binding/group/instance/version/status/transactionId/time
상세	desired/actual version+checksum, target/health, last event/error
Command	해당 화면 operation 만; expected version/checksum + idempotency
Response loss	instance actual snapshot/connection-test operation 조회
Partial	ACK/NACK/UNKNOWN per instance
Recovery	LKG rollback/reapply/reconcile
Audit	publish/change/test requester/approver/reason/result

15.2 /gateway-servers

항목	내용
목적	연동 server group/member
검색	route/binding/group/instance/version/status/transactionId/time
상세	desired/actual version+checksum, target/health, last event/error
Command	해당 화면 operation 만; expected version/checksum + idempotency
Response loss	instance actual snapshot/connection-test operation 조회
Partial	ACK/NACK/UNKNOWN per instance
Recovery	LKG rollback/reapply/reconcile
Audit	publish/change/test requester/approver/reason/result

15.3 /gateway-groups

항목	내용
목적	group topology
검색	route/binding/group/instance/version/status/transactionId/time
상세	desired/actual version+checksum, target/health, last event/error
Command	해당 화면 operation 만; expected version/checksum + idempotency
Response loss	instance actual snapshot/connection-test operation 조회
Partial	ACK/NACK/UNKNOWN per instance
Recovery	LKG rollback/reapply/reconcile
Audit	publish/change/test requester/approver/reason/result

15.4 /gateway-routes

항목	내용
목적	binding/route state
검색	route/binding/group/instance/version/status/transactionId/time
상세	desired/actual version+checksum, target/health, last event/error
Command	해당 화면 operation 만; expected version/checksum + idempotency
Response loss	instance actual snapshot/connection-test operation 조회
Partial	ACK/NACK/UNKNOWN per instance
Recovery	LKG rollback/reapply/reconcile
Audit	publish/change/test requester/approver/reason/result

15.5 /gateway-security

항목	내용
목적	security limits
검색	route/binding/group/instance/version/status/transactionId/time
상세	desired/actual version+checksum, target/health, last event/error
Command	해당 화면 operation 만; expected version/checksum + idempotency
Response loss	instance actual snapshot/connection-test operation 조회
Partial	ACK/NACK/UNKNOWN per instance
Recovery	LKG rollback/reapply/reconcile
Audit	publish/change/test requester/approver/reason/result

15.6 /gateway-health

항목	내용
목적	connection test/revalidate
검색	route/binding/group/instance/version/status/transactionId/time
상세	desired/actual version+checksum, target/health, last event/error
Command	해당 화면 operation 만; expected version/checksum + idempotency
Response loss	instance actual snapshot/connection-test operation 조회
Partial	ACK/NACK/UNKNOWN per instance
Recovery	LKG rollback/reapply/reconcile
Audit	publish/change/test requester/approver/reason/result

15.7 /gateway-transactions

항목	내용
목적	transaction/attempt trace
검색	route/binding/group/instance/version/status/transactionId/time
상세	desired/actual version+checksum, target/health, last event/error
Command	해당 화면 operation 만; expected version/checksum + idempotency
Response loss	instance actual snapshot/connection-test operation 조회
Partial	ACK/NACK/UNKNOWN per instance
Recovery	LKG rollback/reapply/reconcile
Audit	publish/change/test requester/approver/reason/result

15.8 /gateway-log-policies

항목	내용
목적	log policy distribution
검색	route/binding/group/instance/version/status/transactionId/time
상세	desired/actual version+checksum, target/health, last event/error
Command	해당 화면 operation 만; expected version/checksum + idempotency
Response loss	instance actual snapshot/connection-test operation 조회
Partial	ACK/NACK/UNKNOWN per instance
Recovery	LKG rollback/reapply/reconcile
Audit	publish/change/test requester/approver/reason/result

15.9 /gateway-apply-status

항목	내용
목적	instance apply history
검색	route/binding/group/instance/version/status/transactionId/time
상세	desired/actual version+checksum, target/health, last event/error
Command	해당 화면 operation 만; expected version/checksum + idempotency
Response loss	instance actual snapshot/connection-test operation 조회
Partial	ACK/NACK/UNKNOWN per instance
Recovery	LKG rollback/reapply/reconcile
Audit	publish/change/test requester/approver/reason/result